Suggested problems: 1-27 odd, 31, 33

Problem 1: Sketch the following vector fields in $\mathbb{R}^2$.

(a) $\vec{F}(x, y) = \langle 2, 3 \rangle$

(b) $\vec{F}(x, y) = \langle y, 0 \rangle$

(c) $\vec{F}(x, y) = \langle x, y \rangle$

(d) $\vec{F}(x, y) = \langle y, -x \rangle$

(e) $\vec{F}(x, y) = \dfrac{x\vec{i} + y\vec{j}}{\sqrt{x^2 + y^2}}$

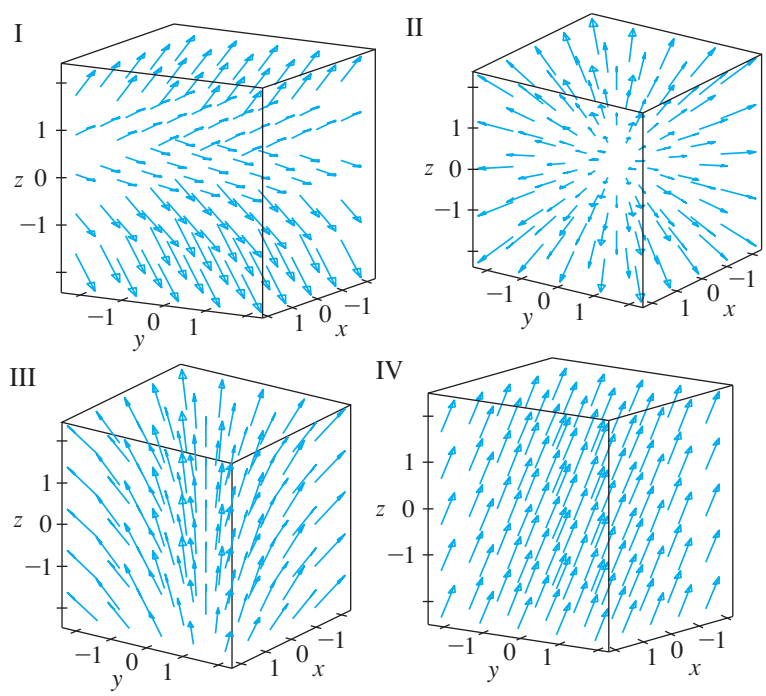
Problem 2: Match the vector fields in $\mathbb{R}^3$ to their formulas.

(a) $\vec{F}(x, y, z) = \langle 1, 2, 3 \rangle$

(b) $\vec{F}(x, y, z) = \langle 1, 2, z \rangle$

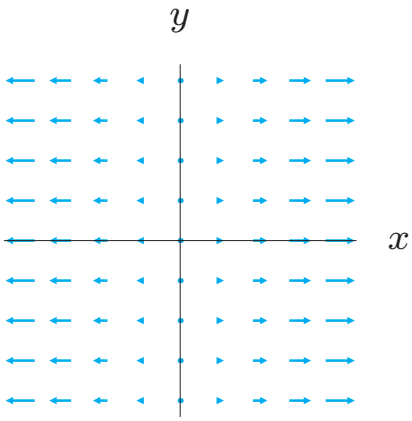
(c) $\vec{F}(x, y, z) = \langle x, y, 3 \rangle$

(d) $\vec{F}(x, y, z) = \langle x, y, z \rangle$

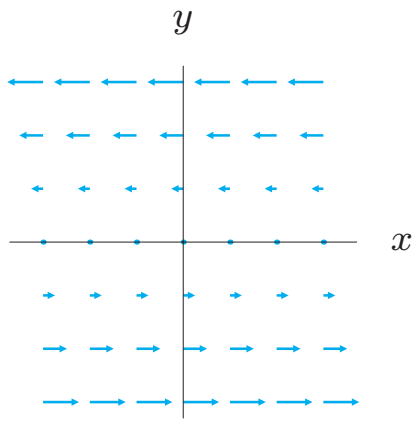


Problem 3: Find formulas for the vectors fields below.

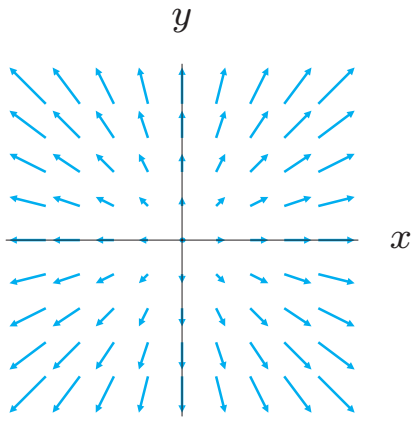
(a)



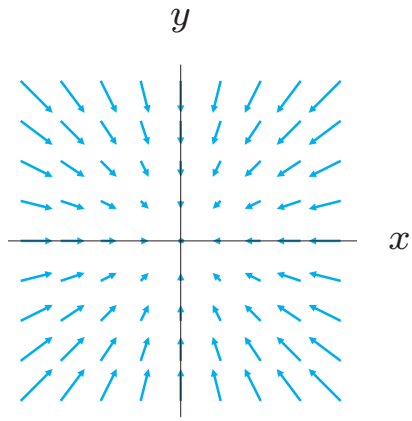
(b)



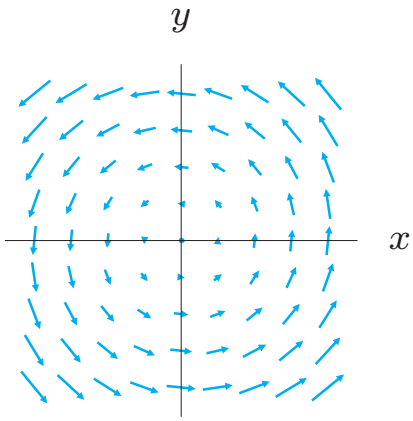
(c)



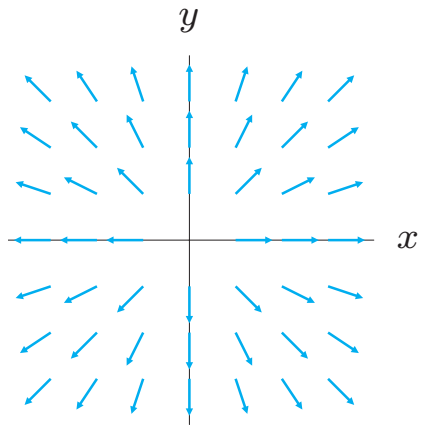
(d)



(d)



(f)



Problem 4: For each description of a vector field in parts (a)-(d), choose one or more of the vector fields (i)-(ix).

(a) Pointing radially outward, increasing in length away from the origin.

(b) Pointing in a circular direction around the origin, remaining the same length.

(c) Pointing toward the origin, increasing in length farther from the origin.

(d) Pointing clockwise around the origin.

(i) $\dfrac{\langle x, y\rangle}{\sqrt{x^2+y^2}}$

(ii) $\dfrac{\langle -y, x\rangle}{\sqrt{x^2+y^2}}$

(iii) $x\vec{i} + y\vec{j}$

(iv) $-x\vec{i} - y\vec{j}$

(v) $\langle -y, x\rangle$

(vi) $\langle y, -x\rangle$

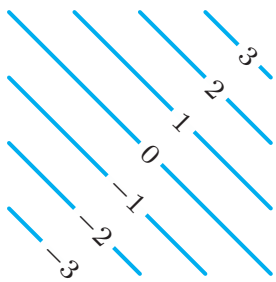
(vii) $\langle y, x\rangle$

(viii) $\dfrac{\langle x, y\rangle}{(x^2+y^2)^{3/2}}$

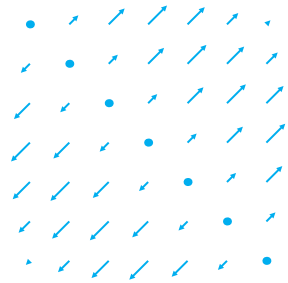
(ix) $-\dfrac{\langle x, y\rangle}{(x^2+y^2)^{3/2}}$

Problem 5: Match the contour plots in (a)-(d) with the gradient fields in (i)-(iv).

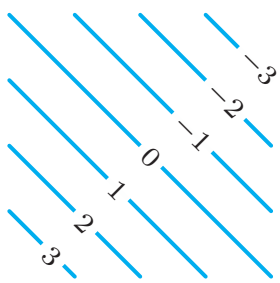
(a)



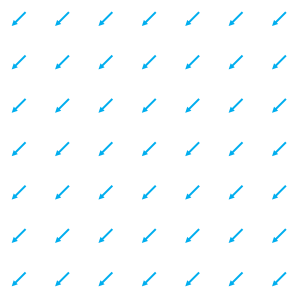
(i)



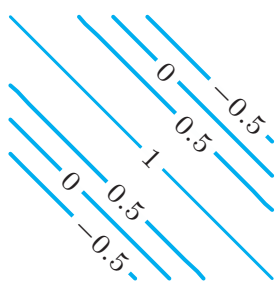
(b)



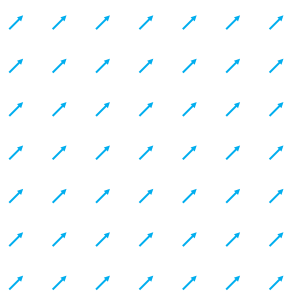
(ii)



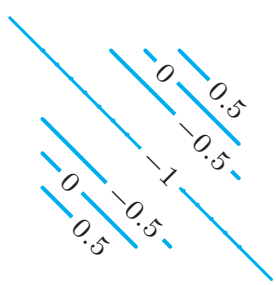
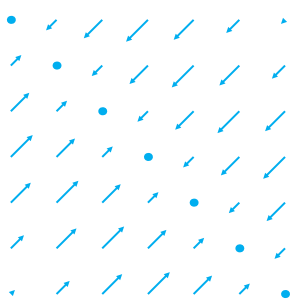
(c)



(iii)



(d)



(iv)